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Original Article

How PCA helps multi-criteria decision making for feature selection: A feature fusion approach in bioinformatics and gene expression data

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ABSTRACT

In high-dimensional data analysis, unsupervised feature selection plays a crucial role in enhancing model interpretability and reducing computational cost. While Principal Component Analysis (PCA) and Multi-Criteria Decision-Making (MCDM) methods such as MOORA have individually been employed for dimensionality reduction and feature evaluation, their combined use remains underexplored in the context of unsupervised feature selection. This study proposes a structured hybrid approach that integrates PCA for extracting dominant components and MOORA for ranking original features based on their alignment with those components. Unlike traditional methods that rely on a single criterion or lack interpretability, our fusion method incorporates multiple decision metrics in a unified framework. The proposed approach is evaluated on both bioinformatics datasets and diverse real-world applications, demonstrating consistent improvements in classification accuracy and feature reduction compared to standalone PCA, MOORA, and other baseline techniques. These results suggest that the synergy between PCA and MCDM can provide a more robust and generalizable strategy for unsupervised feature selection across domains.

1. Introduction

With the explosive increase in data generation across diverse domains—including medical diagnostics, industrial automation, environmental monitoring, and intelligent systems—analysts now face unprecedented challenges in managing large-scale, high-dimensional datasets. These datasets often contain redundant, irrelevant, or noisy information that impedes reliable inference, efficient computation, and robust decision-making [1–6]. As a result, there is a growing demand for advanced unsupervised techniques capable of efficiently handling such datasets while preserving interpretability and general applicability. Recent research has demonstrated that hybrid computational frameworks, which combine dimensionality reduction with principled ranking mechanisms, can address these challenges and offer scalable performance across multiple real-world contexts [4,7,8]. This paper contributes to this evolving area by proposing a general-purpose unsupervised approach designed to enhance feature interpretability and selection without relying on labeled data.

High-dimensional data is increasingly prevalent across a range of real-world applications, including text classification, image recognition, genomics, and social network analysis. However, managing such data presents significant challenges, such as the curse of dimensionality, increased storage and computational requirements, feature redundancy, and the presence of noisy or irrelevant features. To address these issues, dimensionality reduction has emerged as a critical and effective solution [9–13].

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Dimensionality reduction methods are generally categorized into two approaches: feature extraction (FE) and FS. FE focuses on transforming raw data into new feature representations that improve model performance while reducing dimensionality. Techniques such as PCA, Independent Component Analysis (ICA), and Non-negative Matrix Factorization (NMF) are commonly employed to identify linear or nonlinear combinations of original attributes that capture the most relevant information or variance in the data. In contrast, feature selection involves identifying and selecting a subset of the most important original features to enhance model efficiency and performance. It is important to note that FS is distinct from sample selection, which entails choosing a subset of data samples to reduce computational cost or focus analysis [14–17].

This article aims to present a novel FS method. Accordingly, this section introduces various existing FS techniques. FS methods can be broadly categorized into three types based on the availability of class labels: supervised, semi-supervised, and unsupervised. Supervised FS methods identify discriminative features by leveraging the relationship between features and known labels. Semi-supervised approaches are designed to select informative features when only a limited number of labeled samples are available. In contrast, unsupervised FS relies solely on the intrinsic properties of the data, without the need for labeled information. Given that manual labeling is often time-consuming, expensive, and impractical in many industrial applications, unsupervised feature selection has become particularly important for real-world use cases [18–20].

Unsupervised FS techniques have garnered significant attention from the scientific community, driven by recent technological advancements and the growing volume of unlabeled data generated by applications such as text mining, bioinformatics, image retrieval, social media, and intrusion detection. These techniques offer two major advantages. First, unlike supervised FS methods—which may struggle with previously unseen classes—unsupervised approaches are more objective and perform effectively in scenarios where no prior information is available. Second, they help reduce the risk of overfitting. Furthermore, the process of labeling training samples is often time-consuming, costly, and requires domain-specific expertise, making unsupervised FS particularly valuable in real-world applications [19–24].

In many fields, the availability of large, high-dimensional datasets has become increasingly common. However, working with such data presents significant challenges, including increased memory consumption, longer computational time, and a higher risk of overfitting due to the large number of features. These issues often hinder efficient analysis and model performance. PCA is a widely adopted dimensionality reduction technique designed to address such challenges. Rather than selecting a subset of original features, PCA transforms the data into a new coordinate system defined by uncorrelated variables known as principal components. These components are ordered based on the amount of variance they capture from the original data. PCA is particularly valued for its ability to reduce complexity while preserving the essential structure and variability within the data. It also enhances interpretability, facilitates visualization in lower-dimensional spaces, and improves the efficiency of subsequent learning algorithms, especially in high-dimensional environments such as image analysis, bioinformatics, and finance [25].

Real-world decision-making scenarios are often complex and inconsistent, making it challenging to find an optimal solution by analyzing only a single factor. In fact, focusing on just one criterion can lead to a local optimum rather than a global one. A more effective approach involves simultaneously considering all relevant indicators of the problem. MCDM is a field dedicated to developing and applying methods and tools to address complex decision-making challenges. Generally, MCDM aims to evaluate multiple options based on the perspectives of a group of decision-makers or several distinct criteria. In recent decades, researchers have made significant efforts to develop efficient solutions for FS, which is an NP-hard problem. Recently, many FS challenges have been successfully tackled using MCDM techniques, leading to notable progress in this domain [25–28].

Feature selection in unsupervised high-dimensional settings remains a challenging problem, particularly in domains where interpretability of original features is essential, such as bioinformatics, text analysis, or medical diagnostics. PCA is a popular dimensionality reduction method that transforms features into orthogonal principal components. However, it is primarily a feature extraction technique that discards the original feature space, making it unsuitable for applications requiring domain-specific interpretation of selected features (e.g., identifying key genes or meaningful terms) [29].

To address this limitation, we propose a novel framework that integrates PCA with MCDM techniques to perform interpretable, unsupervised feature selection. Specifically, we use PCA to compute reconstruction error for each original feature—quantifying how well each feature is preserved in the reduced representation—and then use MOORA, a well-established MCDM method, to rank and select features based on multiple evaluation criteria. Unlike traditional PCA-based ranking approaches that rely on eigenvalues or loadings, our method leverages reconstruction error as a more interpretable and discriminative measure.

FS methods based on MCDM proposed to date have primarily focused on combining basic feature selection criteria. In contrast, this paper is the first to integrate MCDM with the feature extraction process, introducing a novel FS approach based on the fusion of principal components [30,31]. Among MCDM methods, MOORA is chosen due to its simplicity, low computational complexity, and its ability to handle both benefit and cost criteria without requiring subjective weighting or complex normalization. While MCDM techniques such as TOPSIS and VIKOR have also been explored for feature selection, our approach is distinct in how it integrates unsupervised PCA-driven error metrics into a multi-criteria decision framework. As we show through empirical comparisons, this fusion enables our method to retain the strengths of PCA while overcoming its key limitation—loss of feature identity.

To the best of our knowledge, this is the first study to employ PCA reconstruction error in combination with MOORA for unsupervised feature selection. We demonstrate the effectiveness and generalizability of the proposed method across multiple datasets, including gene expression, text, and image domains. The results highlight improvements in both classification performance and interpretability, establishing our method as a theoretically grounded and practically useful alternative to existing feature selection techniques.

The contributions of this paper are summarized as follows:

- Innovative Use of PCA: While PCA is traditionally used for unsupervised feature extraction, this paper repurposes this method for unsupervised feature selection.
- Reconstruction-Based Feature Differentiation: By reconstructing the data using each PCA eigenvector, achieved a near-identical representation of the overall data. This approach enables us to identify features that exhibit greater variation among samples, which forms the basis of the proposed method.
- Multi-Criteria Analysis via MCDM: By leveraging MCDM, the proposed model evaluates features based on multiple criteria simultaneously, rather than relying on a single evaluation metric. This allows us to account for the influence of each eigenvector on the data's structure.
- Hyperparameter-Free Eigenvector Selection: The derived formula to determine the number of eigenvectors for each dataset, considering the logarithmic decrease in their impact. This eliminates the need for hyperparameters, making the proposed algorithm more adaptable.

The organization of this paper is as follows: Section 2 reviews the performance of recently published unsupervised feature selection methods, as well as applications of PCA and MCDM. Section 3 provides

a detailed explanation of the PCA and MCDM techniques. Section 4 presents the step-by-step phases of the proposed method. Section 5 provides experimental results and compares the performance of the proposed method with other unsupervised FS techniques using real-world datasets. Finally, the paper concludes with an evaluation of the method's effectiveness based on the results discussed.

2. Literature review

This section reviews several research studies closely related to the topic under investigation, including articles on unsupervised feature selection, MCDM-based methods, and PCA-based approaches.

2.1. Unsupervised feature selection methods

Moslemi and Jamshidi [32] introduced an autoencoder-based method for FS aimed at addressing challenges such as high dimensionality, overfitting, and computational cost. They emphasize the significance of both the encoding and decoding phases in the FS process and propose a novel model that preserves the data's topological structure using a Laplacian graph. The study underscores the vital role of FS in various application domains, including healthcare and finance.

DMDSL [33] integrates dual manifold learning with dual graph latent representation. This method considers data correlations, captures local structures, and jointly guides the learning process for both feature transformation and pseudo-label matrices. It effectively preserves local information while minimizing semantic loss and noise interference, utilizing a unified framework to enhance FS performance. Furthermore, DMDSL incorporates non-negative constraints to improve optimization and enhance feature discrimination through iterative refinement.

Hassanieh and Chehade [34] introduced a novel autoencoder-based unsupervised system called Selective Deep Autoencoder (SDAE). This system leverages the capabilities of deep autoencoders for both feature subset selection and imputation, enabling it to naturally identify a subset of relevant features that effectively represent the entire feature space.

Lin et al. [35] presented an efficient unsupervised FS method that combines orthogonal basis clustering with adaptive graph regularization. This approach achieves strong cluster separation while preserving local data structures and employs an effective alternative optimization technique to enhance performance.

A novel unsupervised FS technique called Graph Learning Unsupervised Feature Selection (GLUFS) [36] integrates two traditionally separate processes—similarity matrix construction and feature selection—into a unified framework. GLUFS aims to preserve the manifold structure of the data while generating a new similarity matrix directly from the original dataset

Taha et al. [37] introduced a versatile multiple association measure designed for mixed datasets, along with a novel FS algorithm that leverages these associations. The core idea is that an effective feature set should demonstrate high diversity with minimal interdependence. Their approach formulates the FS problem as an optimization task aimed at identifying features with the lowest degree of association. To this end, they propose two FS algorithms: the Naive Feature Selection Algorithm (NFSA) and the Greedy Feature Selection Algorithm (GFSA), both grounded in their multiple association measure.

Building on the principles of Uniform Manifold Approximation and Projection (UMAP) and nonlinear sparse learning through Feature-Wise Kernelized Lasso, Wang et al. [38] proposed the Multi-Cluster Unsupervised Nonlinear Feature Selection (MUNFS) method. This innovative approach addresses complex nonlinear interactions and improves the representation of high-dimensional data during dimensionality reduction. By employing the Hilbert–Schmidt Independence Criterion (HSIC) to quantify nonlinear dependencies and applying regularization to enforce feature sparsity, MUNFS effectively preserves both local and global data structures while capturing the inherent topology of the data.

Zhou et al. [39] proposed a bi-level FS ensemble method designed to enhance clustering performance. The approach incorporates two levels of ensembling: feature-level and clustering-level. Initially, a feature-level ensemble is performed, followed by the generation of a consensus clustering result based on the base FS outcomes using self-paced learning. This consensus result subsequently guides and refines the FS process, creating a feedback loop that improves the overall effectiveness of the method.

Li et al. [40] focused on unsupervised FS by learning the reconstruction function directly from the data. They introduced a novel reconstruction-based unsupervised FS framework, called REFS, which integrates the reconstruction function learning process into the FS procedure to improve its effectiveness.

General Adaptive Unsupervised Feature Selection with Auto-weighting (GAWFS) model [41] takes a distinct approach by identifying key features directly from raw data to form well-separated clusters, rather than reducing the data to a compressed representation. It leverages adaptive graph learning and non-negative matrix factorization to identify critical clustering features. Additionally, the model employs a sparse weighting matrix to effectively determine the most relevant features.

2.2. PCA-based methods

This paper [42] presents a personalized PCA method, referred to as Performance-Based Analytics, for identifying low-dimensional common and unique features within high-dimensional data collected from distributed edge devices in IoT systems. The model accounts for the unique characteristics of each client and utilizes orthonormal principal components to explain variations in the observations.

The research [43] presents an innovative approach that combines environmental data with financial metrics to predict stock values, employing a hybrid PCA-GRU-LSTM model. This model integrates PCA, Gated Recurrent Units (GRU), and Long Short-Term Memory (LSTM) networks to effectively capture both dimensionality-reduced input features and complex temporal dependencies.

The research [44] improves survival prediction in severe accidents by integrating PCA, hyperparameter optimization, resampling techniques, and the Edited Nearest Neighbors (ENN) algorithm. Utilizing seven machine learning algorithms, the model demonstrates enhanced accuracy and generalization performance.

This paper [45] proposes the integration of modern machine learning algorithms with dimensionality reduction techniques. Specifically, it combines XGBoost with optimized PCA to enhance data structuring and accelerate processing, particularly when working with large datasets.

This paper [46] examines the application of PCA to diminish embedding dimensionality, thus alleviating computing constraints without significant accuracy degradation.

2.3. MCDM-based methods

In [47], a q-rung orthopair hesitant fuzzy MCDM approach is employed to extend the CODAS technique for multi-label FS. This represents the first extension of the CODAS algorithm that incorporates information fusion to generate a decision matrix.

The study [48] introduces AE-MCDM, a novel unsupervised FS technique that integrates MCDM with autoencoder-based feature extraction. By leveraging deep representation learning, it enhances the accuracy and effectiveness of feature evaluation.

The paper [49] describes a heart disease diagnosis system that integrates MCDM, FS methods, and artificial neural networks. To accurately identify fifteen diagnostic categories, the system employs three neural networks, five FS techniques, Particle Swarm Optimization, an integrated weighting mechanism, and an enhanced ELECTRE III method.

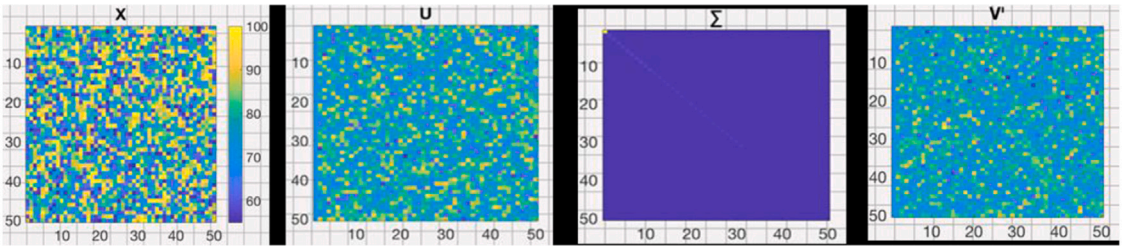


Fig. 1. Decomposition from full single value to reduced singular value. The matrix was produced at random.

Within an MCDM framework, the paper [50] proposes a cost-sensitive multi-label FS technique that balances feature relevance and associated costs. It ranks features based on mutual information and Fermatean fuzzy sets to evaluate both relevance and cost. Experimental results demonstrate that this approach effectively balances competing objectives to achieve globally optimal outcomes, outperforming existing methods.

This study [51] offers a comprehensive overview of FS methodologies, evaluation criteria, and underlying rationales. It also examines the advantages and limitations of advanced FS techniques, including those based on MCDM.

3. Fundamental concepts

3.1. Principal components analysis (PCA)

PCA is a widely used technique for reducing the dimensionality of high-dimensional feature vectors. In many cases, it can significantly decrease computation time with little to no impact on recognition accuracy. Moreover, PCA is effective in identifying an orthogonal basis set for highly redundant features, such as raw pixel values in small image patches [52,53].

The primary objective of PCA is to identify patterns in data and uncover the relationships between variables. Dimensionality reduction is only meaningful when there is a significant correlation among the variables. Essentially, PCA aims to find the directions of greatest variance in high-dimensional data and project the data onto a lower-dimensional subspace while retaining most of the information. Mathematically, PCA is defined as an orthogonal linear transformation that repositions the data into a new coordinate system. In this system, the first coordinate (referred to as the first principal component) captures the highest variance in the data, the second coordinate captures the second-highest variance, and so forth [29]. Assume that X is:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix} \tag{1}$$

There is an orthogonal matrix $U_{n\times n}$ and $U_{p\times p}$ such that $U^T XV = \Sigma$ for any matrix $X_{n\times p}$. Where sigma, $\Sigma = diagonal\ (\sigma_1,\ \sigma_2,\ \sigma_3,\ ....,\sigma_n)_{n\times p}$ $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \ldots \geq \sigma_n \geq 0$. The columns of U and V are referred to as the left and right singular vectors of X, respectively, and the σ_i's are referred to as the singular values of X. $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \ldots \geq \sigma_d \geq \sigma_{d+1} = \sigma_{d+2} = \sigma_{d+3} = \sigma_{d+4} = 0$ for n singular values where others are zero and d is the number of positive singular values. Our initial matrix X now has the rank d (Fig. 1). The span of $u_1,\ u_2,\ u_3,\ \ldots,\ u_d$, which equals d, is the range of X. The dimension of X was $n \times p$ at first, but it is now $n \times d$. In terms of statistics, the quantity of data is decreased from p to d. $U^T XV = \Sigma$ now becomes $X = U\Sigma V^T$, which means that we had p columns in the original data and now have d columns after SVD [54,55].

With this SVD method, we can now do principal component analysis (PCA). At first, our X resembles a column shape.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_p \tag{2}$$

With covariance K and mean zero. However, our primary component, X_d, has now lowered it. Please take note that $X - E(X)$ should be used for centering if the mean of X is not zero [35].

At this point, the covariance matrix is K, which appears to:

$$K = \begin{bmatrix} cov(x_1, x_1) & \cdots & cov(x_1, x_n) \\ \vdots & \ddots & \vdots \\ cov(x_n, x_1) & \cdots & cov(x_n, x_n) \end{bmatrix} \tag{3}$$

Let U be the $p \times p$ orthogonal matrix in which the vectors $Z = U^T X$ have uncorrelated components. The components of X are the uncorrelated elements of the vector $Z = U^T X$. Therefore, our goal is to determine this U, which can be found using the covariance matrix K' s SVD. $cov(Z) = U^T\ cov(X)U = U^T KU = \Sigma$, to be exact. This SVD may now be compared.

The principal component is therefore $Z_i = U_i^T X$. In further detail, the main components and directions are $(Z_1,\ Z_2,\ Z_3,\ \ldots)$ and $(U_1,\ U_2,\ U_3,\ \ldots)$, respectively. In this case, Z_1 has the most deviation, followed by Z_2 and so forth. Thus, the PCA procedure for the provided data may be reduced to the following, where X represents an independent observation vector $'X'$. The orthogonal matrix $U_{d\times n}$ is obtained by computing the SVD of K after determining the sample covariance $K_{p\times p}$. Next, the first column of the matrix U was selected to define the primary directions. Consequently, the transformed feature vector was obtained as:

$$Z_{d\times 1} = U_{d\times n}^T X_{n\times 1} \tag{4}$$

It should be noted that d is significantly smaller than n when the dataset contains a high level of correlated features [54].

Also, to reconstruct the original data based on the selected principal components, U must be multiplied by the obtained Z matrix.

$$reconstructed\ X = (UZ)^T \tag{5}$$

With this method, the amount of variance in the data that is preserved by PCA can be easily examined [49].

Algorithm 1 presents the sequential process of the PCA algorithm.

3.2. Multi criteria decision making (MCDM) method and multi-objective optimization on the basis of ratio analysis (MOORA) algorithm

Brauers et al. [56] proposed Multi-Objective Optimization on the basis of Ratio Analysis (MOORA), which integrates the ratio system with the reference point technique. Prior to executing the MCDM process, a weight vector and a decision matrix must be established. Accordingly, in the MOORA method, the ratings x_{ij} of m alternatives

Algorithm 1 PCA Algorithm

Require: Input: $n \times p$ data matrix X, number of reduced dimensions d
Ensure: Output: $n \times n$ eigen vectors calls U and $1 \times n$ eigen vectors call $\sum$

1: Compute the mean of each column of X:

$$X_{\text{mean}} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

2: Center the data:

$$\text{Centered_X} = X - X_{\text{mean}}$$

3: Compute the covariance matrix:

$$K = Covariance\ of\ Centered\ X$$

4: Perform Singular Value Decomposition (SVD) on K:

$$U \Sigma V^T = return\ \text{SVD}(K)\ on\ K$$

5: U, $\sum$ are eigen vectors and eigen values :

Algorithm 2 MOORA algorithm

Require: Input: $m \times n$ decision matrix called X and weight vector called $W = [w_1, w_2, w_3, \ldots, w_n]$
Ensure: Output: Ranking of alternatives

1: **for** $i = 1$ to m **do**
2: **for** $j = 1$ to n **do**
3: Compute normalized values:

$$\text{normalized_x}_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} (x_{ij})^2}}$$

4: **end for**
5: **end for**
6: **for** $i = 1, 2, 3, \ldots, m$ **do**
7: Compute ranking scores:

$$U_i = \sum_{j=1}^{n} w_j \cdot \text{normalized_x}_{ij}$$

8: **end for**
9: Rank the features based on U_i in descending order

with respect to n criteria are utilized to construct the decision matrix (X) as follows:

$$X_{m \times n} = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{1m} & \cdots & x_{mn} \end{bmatrix}_{m \times n} \tag{6}$$

Each criterion may occasionally have varying degrees of impact on the outcomes of an MCDM. Consequently, the following is the definition of a weight vector that displays the weighting of criteria:

$$W = [w_1,\ w_2,\ w_3,\ \ldots\ w_n] \tag{7}$$

First the decision matrix is normalized using the following equation:

$$normalaized_x_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} (x_{ij})^2}} \tag{8}$$

where i and j are the alternative number and criterion number, respectively. The normalization matrix is constructed as follows:

$$normalaized_X = \begin{bmatrix} normalaized_x_{11} & \cdots & normalaized_x_{1n} \\ \vdots & \ddots & \vdots \\ normalaized_x_{n1} & \cdots & normalaized_x_{nn} \end{bmatrix} \tag{9}$$

where g is the number of criteria that are beneficial, and $(n - g)$ is the number of criteria that are not beneficial. All of the factors in our proposed algorithm are advantageous. Therefore, the following formula determines the utility of ith option:

$$U_i = \sum_{j=1}^{g} w_j \ .\ normalaized_x_{ij} \tag{10}$$

4. Proposed method

This section presents the motivation behind the effectiveness of the proposed method and provides a comprehensive description of its formulation.

4.1. Motivation

The proposed FS algorithm integrates PCA with a MCDM method, specifically MOORA, to identify and select informative and diverse features. PCA, a widely adopted dimensionality reduction technique, transforms the original data into an orthogonal basis in which each principal component captures a distinct portion of the data's covariance structure. Each component reflects a unique direction of maximum variance, effectively isolating specific patterns of correlation among features. This property is exploited to ensure that the selected features align with different covariance patterns, thereby minimizing redundancy within the feature set.

In this algorithm, the reconstruction error is employed as a criterion to evaluate each feature's contribution to the overall data structure. By reconstructing the data matrix using individual eigenvectors (principal components), the Euclidean distance between the original matrix and the reconstructed matrix is computed for each feature. This distance quantifies the information loss when only the variance explained by a specific principal component is considered. Features with lower reconstruction errors are better aligned with the information captured by the corresponding principal component.

To perform feature selection, the MOORA method is applied to aggregate reconstruction errors across all principal components. Within the MOORA framework, features are treated as alternatives, and the reconstruction errors corresponding to different principal components serve as evaluation criteria. These errors are normalized to ensure comparability across criteria. The method balances trade-offs by simultaneously minimizing reconstruction error and maximizing diversity in the selected features. As a result, the final feature set consists of features that capture distinct covariance structures while contributing meaningful information to the data representation.

The theoretical justification for this approach is grounded in the properties of PCA and MCDM. First, PCA's orthogonal transformation ensures that principal components are uncorrelated, so features aligned with different components capture diverse and non-redundant information. Second, minimizing reconstruction error ensures that the selected features retain significant variance and meaningful structure from the original data. Lastly, the MOORA method balances these objectives by identifying features that contribute both to accuracy and diversity. Together, these properties guarantee that the algorithm selects high-quality features—those that are both representative of the data's underlying structure and minimally redundant. The effectiveness of the algorithm can be further validated by applying it to synthetic datasets with known covariance structures and assessing its impact on downstream tasks such as classification or regression.

4.2. Proposed algorithm

This section provides a detailed description of the proposed algorithm. The method is presented both as pseudocode in Algorithm 3 and as a flowchart in Fig. 2.

To select data features using the proposed method, the process begins by calculating the mean of each feature vector. This mean is subtracted from the original data, and the resulting data is used to

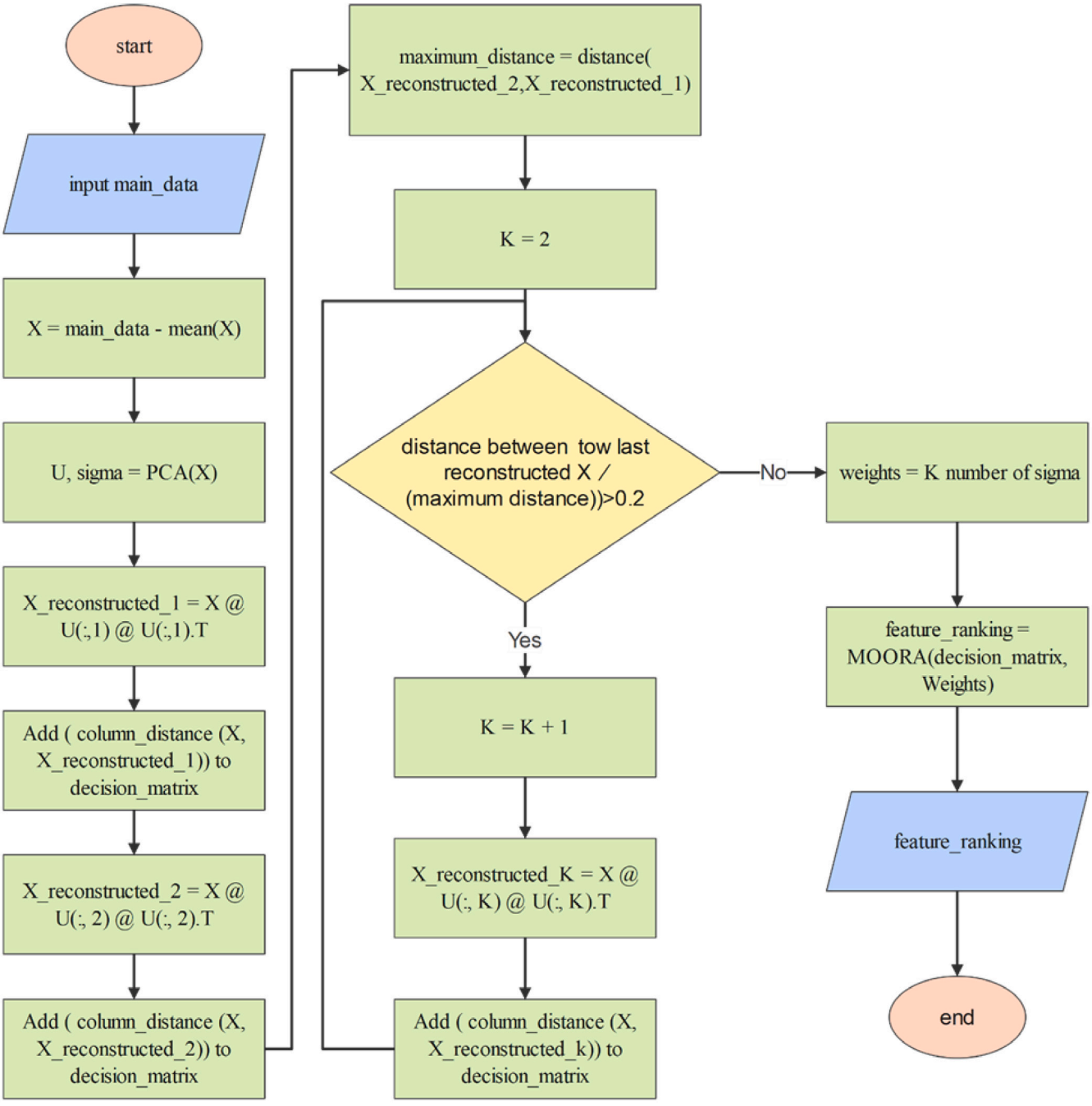


Fig. 2. Flowchart of proposed method.

replace the original data. This procedure continues iteratively until the algorithm reaches its conclusion.

Then the covariance matrix will be calculated as follows:

$$C = \begin{bmatrix} cov(x_1, x_1) & \cdots & cov(x_1, x_n) \\ \vdots & \ddots & \vdots \\ cov(x_n, x_1) & \cdots & cov(x_n, x_n) \end{bmatrix} \tag{11}$$

Where x_i is values of ith feature.

Then, calculate SVD of this covariance matrix, called C. To obtain the vectors U, and values of Σ. Which are arranged such that the ith value of Σ vector corresponds to the ith vector of the matrix U and also the ith value of Σ vector is larger than or equal to the $(i+1)$th value and less than or equal to the $(i-1)$th value.

Then, we take the first vector of the U matrix and multiply it by the data (the data obtained at the beginning by subtracting the mean vector of each feature). Then, we multiply the result of this multiplication again by the first vector of the U matrix to rebuild the data matrix based on this vector. Like Definition 1.

Definition 1. here is how to reconstruct first data from a vector of U matrix:

$$X_reduced = X.u_i \tag{12}$$

$$X_{reconstruction} = X_{reduced} \times u_i^T \tag{13}$$

Now calculate the Euclidean distance with Definition 2 of each feature of the original data and each feature created and store the obtained distances in a column of the decision matrix.

Definition 2. Euclidean distance between two vector $a, b \in R^n$ is calculated by below formula:

$$distance(a,b) = \sqrt{\sum_{i=1}^{n} (a_i - b_i)^2} \quad (14)$$

The same procedure must be repeated using the second column vector of the matrix U. Subsequently, the Euclidean distance is calculated between the two reconstructed data matrices: one obtained based on the first vector of U, and the other based on the second vector. This distance is referred to as the maximum distance and is formally computed according to Definition 3:

Definition 3. Euclidean distance between two matrix $a, b \in R^{n \times m}$ is calculated by below formula:

$$distance(a,b) = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{m} (a_{ij} - b_{ij})^2} \quad (15)$$

The aforementioned steps are repeated for the remaining column vectors of matrix U until the ratio of the Euclidean distance between the reconstructed matrix at the current step and that of the previous step, divided by the maximum distance (as defined earlier), falls below 0.2. Satisfying this condition indicates that a sufficient number of criteria have been identified for constructing the decision matrix. Therefore, there is no need to evaluate reconstructed matrices for all vectors in matrix U, as beyond this point, the differences become negligible and no longer contribute significantly to the decision-making process. Although Euclidean distance is used in our method to quantify reconstruction error between original and projected features, we recognize that this choice may be sensitive to data scale and does not account for feature correlations. For datasets with high dimensionality or multicollinearity, such as gene expression data, Mahalanobis distance may provide a more nuanced measure by incorporating the data's covariance structure. Future versions of this approach may explore the impact of alternative distance metrics on feature selection outcomes.

To proceed with feature selection using the decision matrix constructed in the previous steps, each column of the decision matrix is first normalized according to Eq. (8), yielding a normalized decision matrix. To make informed decisions based on this matrix, weighting coefficients are introduced to reflect the relative importance of each decision criterion. These coefficients are derived from the singular values in the diagonal matrix $\sum$, where larger singular values correspond to greater contributions of the associated vectors to the reconstruction of the original data.

The eigenvalues obtained from the PCA were employed to assign weights to the decision vectors, as each eigenvalue reflects the relative importance of its corresponding eigenvector in reconstructing the original data. Specifically, an eigenvalue of zero indicates that the associated eigenvector contributes no information to the reconstruction, while larger eigenvalues correspond to greater contributions. Consequently, using eigenvalues as weighting factors to evaluate the significance of each eigenvector is both theoretically sound and practically appropriate. This approach is particularly advantageous because it directly quantifies the reconstruction capacity of each principal component, making it a meaningful criterion for weighting within the decision-making process.

The coefficients were obtained by first extracting the corresponding singular values from vector $\sum$ associated with the selected column vectors of matrix U Each of these extracted values was then divided by the sum of all selected singular values. This procedure is formalized in Definition 4.

Definition 4. here is how to obtained coefficients of the decision matrix:

$$coefficients_of_decision_matrix = \frac{w_i}{\sum_{i=1}^{n} w_i} \quad (16)$$

Finally, by applying these coefficients to the decision matrix and summing the weighted decision values for each feature, the overall importance of each feature is determined. The features are then ranked in descending order of importance, and any desired number of top-ranked features can be selected accordingly. The complete procedure of the proposed method is illustrated in Algorithm.3 and Fig. 2.

Algorithm 3 Proposed Algorithm

Require: *Input* $n \times p$ data matrix `main_data`
Ensure: *Output* Ranking of features
$X_{mean} \leftarrow$ column-wise mean of `main_data`
$X \leftarrow$ `main_data` $- X_{mean}$
$(U, \Sigma) \leftarrow$ PCA(X)
`reconstructed_first` $\leftarrow (X \times U[:,1]) \times U[:,1]^T$
Add distance between each column of `reconstructed_first` and X to `decision_matrix`
`reconstructed_second` $\leftarrow (X \times U[:,2]) \times U[:,2]^T$
Add distance between `reconstructed_second` and X to `decision_matrix`
`max_dist` $\leftarrow$ distance(`reconstructed_first`, `reconstructed_second`)
$K \leftarrow 2$
while $\frac{\text{distance between last two reconstructed } X}{\text{max_dist}} > 0.2$ **do**
  $K \leftarrow K + 1$
  `reconstructed_x` $\leftarrow (X \times U[:,K]) \times U[:,K]^T$
  Add distance between `reconstructed_x` and X to `decision_matrix`
end while
`weights` $\leftarrow \Sigma[1:K]$
`rank_features` $\leftarrow$ MOORA(`decision_matrix`, `weights`)

4.3. Datasets

The real-world datasets used to evaluate the effectiveness of the proposed method are introduced in this subsection. To demonstrate its performance, the method was applied to the datasets listed in Table 1, each of which is briefly described therein:

The Colon dataset includes gene expression data from 62 colon tissue samples: 40 tumor samples and 22 normal samples. About 6,500 genes were analyzed, with the 2,000 genes showing the most variation selected for further analysis. This dataset is a benchmark for testing feature selection and classification algorithms in high-dimensional spaces and is one of the first uses of oligonucleotide arrays in cancer research [57].

The Golub dataset includes gene expression data were obtained from 72 leukemia samples, comprising 47 samples of acute lymphoblastic leukemia (ALL) and 25 samples of acute myeloid leukemia (AML). Using Affymetrix oligonucleotide microarrays, the expression levels of 7129 genes were measured for each sample. These data were normalized and analyzed to identify molecular patterns distinguishing the two leukemia subtypes. The dataset has become a widely used benchmark for evaluating feature selection and classification methods, particularly in high-dimensional gene expression studies, due to its clear structure and relevance to cancer diagnostics [58].

The khan dataset includes gene expression data were obtained from 63 samples of small, round blue-cell tumors (SRBCTs), including four diagnostic categories: Ewing sarcoma (EWS), rhabdomyosarcoma (RMS), neuroblastoma (NB), and Burkitt lymphoma (BL). These samples were derived from both tumor biopsies and cell lines, with expression levels measured for 6,567 genes using cDNA microarrays. After preprocessing, 2,308 genes were selected for further analysis [59].

The NCI-60 dataset is one of the most widely studied resources in cancer research, particularly for analyzing gene expression profiles across various tumor types. The dataset comprises 64 samples representing 60 distinct cancer cell lines derived from tumors of different tissue origins, including leukemia, melanoma, lung, colon, and

Table 1
The description of datasets.

Dataset	Number of instances	Number of features	Number of classes	Domain
Colon	62	2000	2	Genomics Data
Gulab	72	7129	2	Genomics Data
Khan	63	2308	4	Genomics Data
Nci60	64	6830	14	Genomics Data
Shipp	77	7129	2	Genomics Data
Subramanian	50	10 100	2	Bioinformatics Data
Sorlie	85	456	5	Genomics Data
Su	102	5565	4	Genomics Data
Lung	203	3312	5	Genomics Data
Corel	5000	423	50	Image Data
Madelon	2000	500	2	Artificial
USPS	9298	256	10	Hand Written Image Data

breast cancers. The dataset features measurements for approximately 6,830 genes, which have been systematically profiled using cDNA microarrays. This dataset serves as a benchmark for understanding the molecular mechanisms of cancer and exploring therapeutic targets [59].

The Shipp dataset contains 77 samples of diffuse large B-cell lymphoma (DLBCL) and follicular lymphoma (FL). Gene expression profiles of 7,129 genes were measured for each sample to identify molecular patterns distinguishing these two cancer types [60]. Subramanian's dataset contains 10,100 genes and 50 samples. Gene Set Enrichment Analysis (GSEA) is a powerful analytical method for interpreting gene expression data, focusing on gene sets with common biological function or regulation [61].

The Sorlie study analyzed 85 cDNA microarray experiments to classify breast carcinomas based on gene expression patterns and correlate tumor characteristics to clinical outcome. Results showed distinct subgroups with distinct expression profiles, with poor prognosis for the basal-like subtype [62].

The Su dataset, introduced by Su et al. [63], is a gene expression dataset used for cancer classification, containing 102 samples and 5565 features, labeled with four classes.

The Corel dataset consists of 5000 samples with each represented by 423 features. The dataset originates from the Corel image library and is used in content-based image retrieval and clustering studies. It includes 50 classes, with each class containing 100 samples. These classes represent diverse categories of images, making the dataset a challenging benchmark for clustering and classification algorithms in high-dimensional space [64].

Madelon is an artificial dataset used in the NIPS 2003 feature selection challenge [65], a two-class classification problem with continuous input variables. It consists of 32 clusters of data points grouped in a five-dimensional hypercube, randomly labeled +1 or -1. The dataset includes 20 informative features and 480 distractor features. The dataset is split into training, validation, and test sets, with target values provided for the first two sets [66].

The USPS dataset is a widely used benchmark for machine learning and pattern recognition, particularly for handwritten digit classification. It contains 9,298 grayscale images representing handwritten digits, each size-normalized to 16 × 16 pixels. The dataset is suitable for supervised and unsupervised learning tasks due to its balanced distribution across ten classes. Its compact size and realistic variations in handwriting styles make it a standard benchmark for testing algorithms' robustness and efficiency [67].

5. The experimental results

This section presents the experimental evaluation of the proposed method using real-world datasets.

5.1. Comparison methods

The proposed method is compared with seven state-of-the-art unsupervised FS methods, including AEFS [68], DGUFS [69], FRFSO [70], SCAFS [71], SCEFS [71], RSR [72], and EFS-FMCDM [24]. These approaches were selected for comparison based on the following considerations:

(1) **AEFS**[1]: Selected as a representative of autoencoder-based FS methods. It provides a benchmark for evaluating the performance of the proposed algorithm in comparison to neural network-based unsupervised FS techniques.
(2) **DGUFS**[2]: Chosen due to its graph-based learning mechanism for feature selection. Comparing with DGUFS enables the analysis of the strengths and distinctions of the proposed method in relation to graph-driven approaches.
(3) **FRFSO**[3]: Utilizes a firefly algorithm and serves as a representative of evolutionary feature selection techniques. This comparison facilitates the evaluation of the proposed method against population-based metaheuristic approaches.
(4) **SCAFS, SRCFS, and SCEFS:** These methods were included as recent FS algorithms that incorporate sparse representation and collaborative filtering. A comparative analysis with these approaches highlights the effectiveness and novelty of the proposed algorithm in the context of advanced sparse techniques.
(5) **EFS-FMCDM**: This algorithm integrates fuzzy logic and MCDM in an unsupervised FS setting. It was selected to assess the performance of the proposed method in comparison to an alternative MCDM-based approach.
(6) **RSR**[4]: This algorithm is an unsupervised feature selection method that uses data structure to identify global features, ensuring robustness against noise and redundancy.

To ensure a comprehensive evaluation of the performance, effectiveness, and distinctiveness of the proposed feature selection technique, a comparative analysis with several state-of-the-art methods was conducted. This evaluation highlights the competitiveness of the proposed approach and demonstrates its advantages and improvements over existing advanced techniques in the field.

5.2. Experimental settings

This section outlines the experimental settings and evaluation methodology. To assess the efficiency of the proposed algorithm in comparison with baseline approaches, the K-Nearest Neighbors (KNN)

[1] https://github.com/xiangfasong/AEFS
[2] https://github.com/eeGuoJun/AAAI2018_DGUFS
[3] https://github.com/AISKYEYE-TJU/FRFS0-IVC2016
[4] https://github.com/AISKYEYE-TJU/RSR-PR2015

Table 2
The run-time of algorithms (s).

Dataset	PCA-MCDM	AEFS [68]	DGUFS [69]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]	EFS-FMCDM [24]
Colon	0.7150	0.6390	0.3068	15.183	0.1042	**0.0991**	10.598	0.2543
Golub	38.827	1.5703	1.6473	333.16	**0.7743**	0.7891	253	0.7802
Khan	1.0263	0.6378	0.3085	19.446	**0.1147**	0.1162	13.662	0.2228
Lung	3.6806	0.9785	108.71	48.398	0.3367	**0.3321**	35.007	0.8596
NCI60	32.779	1.5269	1.5198	268.33	0.9058	0.9656	201.03	**0.7223**
Shipp	47.468	1.7557	2.0547	579.38	1.0672	**1.0644**	306.87	0.8507
Sorlie	0.0339	0.0897	0.1567	0.4595	0.0044	**0.0042**	0.2584	0.0837
Su	17.315	1.109	1.6716	142.32	0.4953	**0.4919**	106.91	0.5877
Subramanian	89.568	1.8983	1.4418	716.33	1.1257	1.1564	540.6	**0.9485**
Corel	1.0628	0.4852	390.4	3.2324	**0.0264**	0.0283	21.463	58.499
Madelon	0.9579	0.2849	32.69	1.7189	0.0146	**0.0142**	2.9175	5.4371
Usps	0.4143	0.6762	2775.8	2.4298	**0.0214**	0.0216	65.163	265.85

classification algorithm was employed. KNN was selected due to its high sensitivity to irrelevant features, making it particularly appropriate for evaluating the impact of feature selection. The evaluation process utilized holdout validation, where each dataset was randomly split into 70% for training and 30% for testing. This procedure was repeated 30 times to ensure statistical robustness, and the average classification accuracy across all repetitions was reported [73].

The parameter value for every algorithm in every simulation is set in accordance with the advice in the article that follows. All algorithms' classification performance is determined by the k-nearest neighbor (KNN), which is based on F-score and accuracy measures. The KNN classifier's k parameter, which is set to 5. Various numbers of features are chosen from range $\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$ in order to evaluate the effectiveness of the feature selection methods. A Windows 11 Pro for Workstations-64 bit computer with 32 GB of RAM and an Intel(R) Core (TM) i9-14900HX CPU with 24 3.10 GHz processors is used for all of the experiments. MATLAB R2024a is also used.

Finally, to compare the efficiency of the proposed method and other methods, four criteria were used: accuracy, F-measure, precision, and recall. To explain how to calculate them, it is necessary to first define the following four concepts:

- True Positive (TP): number of positive examples that were correctly classified.
- False Positive (FP): number of negative examples that were incorrectly classified as positive.
- True Negative (TN): number of negative examples that were correctly classified.
- False Negative (FN): number of positive examples that were incorrectly classified as negative

In the following, the four evaluation criteria of the above methods will be explained using the four phrases above in Definition 5 to 8.

Definition 5. Accuracy shows what percentage of all samples (regardless of their class) the model correctly classified and is a general measure of the correctness of the model's performance. It is calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{17}$$

Definition 6. Precision for each class determines what percentage of the samples that the model says belong to that class were actually correct; that is, it measures the level of confidence in the model's predictions for that class. It is calculated as follows:

$$Precision = \frac{TP}{TP + FP} \tag{18}$$

Definition 7. Recall shows what percentage of all the examples that actually belonged to a particular class the model correctly identified, so it shows the model's ability to find examples of that class. It is calculated as follows:

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

Definition 8. The F-measure is a combination of precision and recall, expressed as a harmonic mean, and indicates how successful the model was in simultaneously making correct and complete predictions for each class. It is calculated as follows:

$$F - measure = \frac{TP}{TP + \frac{1}{2}(FN + FP)} \tag{20}$$

To address the issue of class imbalance, we report macro-averaged precision, recall, and F-measure scores, ensuring that all classes, including minority ones, are equally represented in the evaluation metrics.

To statistically validate the performance differences between the proposed method and baseline approaches, the Wilcoxon signed-rank test was applied to the four evaluations metrics; accuracy, precision, recall, and F-measure. As a non-parametric paired test, Wilcoxon is suitable for comparing classifier performance across multiple datasets without assuming normality in metric distributions. The detailed results of this statistical comparison are presented in Tables 3 through 6, each corresponding to one of the evaluation metrics while Table 7 provides an overall win/tie/loss summary. The observed p-values confirm the significance of the improvements achieved by the proposed method across all metrics. This approach to statistical comparison is consistent with recent best practices in machine learning evaluation [63]. The plus symbol (+), which may be found in tables 3 through 6, indicates that the PCA-MCDM performs statistically better than other approaches. In addition, the indications with an equal sign (=) and a negative sign (-) indicate the best and worst performances

5.3. Results

In this section, the experiment results are in the real datasets. The user can determine how many features to select using this technique. The results derived from nine datasets are displayed in Figs. 3 to 6. Accuracy, F-measure, Recall and Precision [6] include four widely-known measures that are employed that defined in Definition 5 to 8 to assess how well feature selection algorithms operate during the learning process. Also, the runtime of algorithms is reported in Table 2. In Figs. 3 to 6, the classification performance as a function of the number of selected features is presented. The X-axis shows the number of top-ranked features used, and the Y-axis shows the corresponding performance. This form allows direct comparison of method performance as feature subset size increases.

In Fig. 3, the results are evaluated based on the classification accuracy defined in Definition 5, which quantifies the proportion of correctly classified samples among all samples. This metric shows how well the proposed method outperforms competing approaches in terms of overall prediction accuracy.

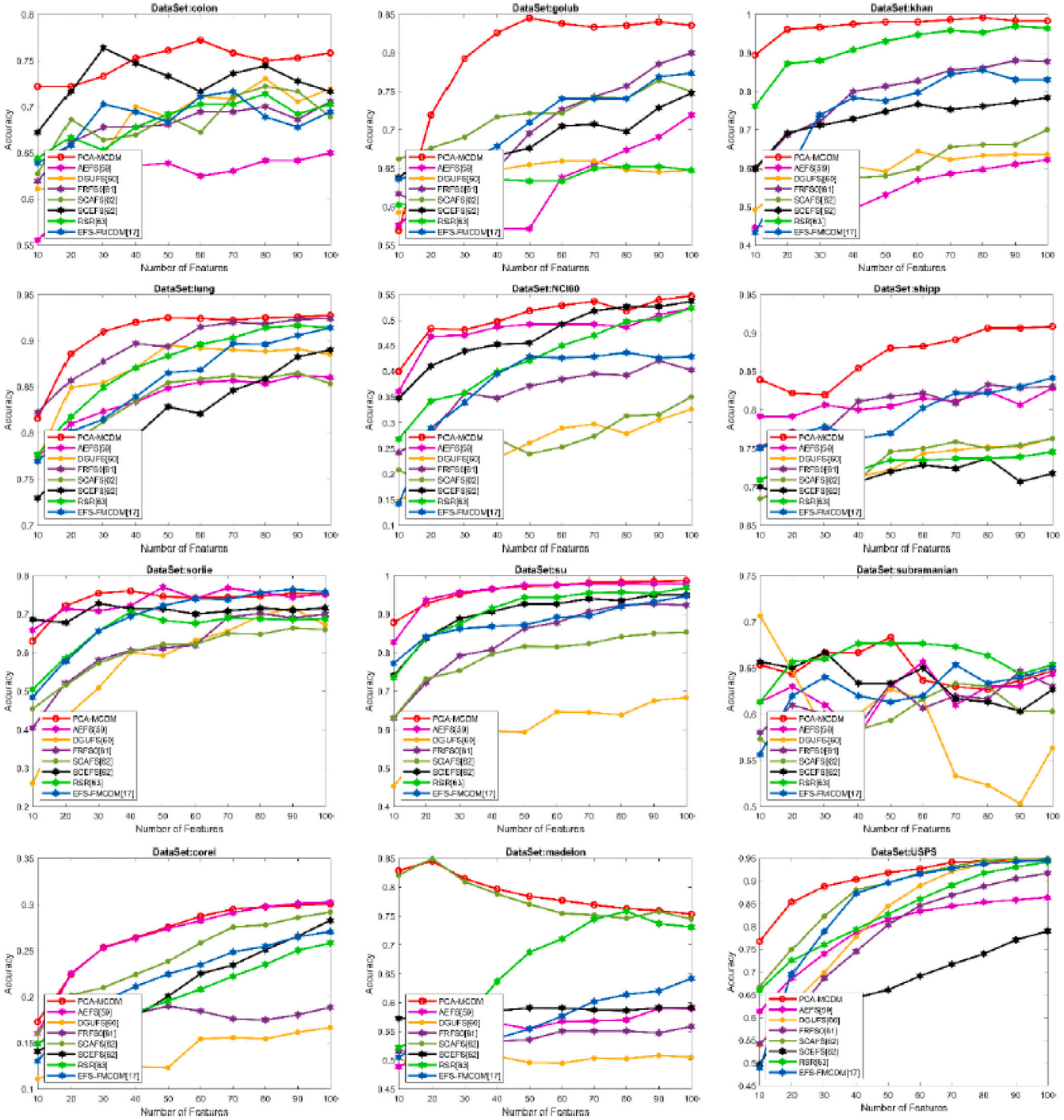


Fig. 3. The results based on classification accuracy.

Fig. 4 shows the performance of the proposed method using the F-measure metric, as introduced in Definition 8, revealing a success of the proposed method competing other methods was in simultaneously making correct and complete predictions for each class.

Fig. 5 presents the performance of the proposed method in terms of precision, as defined in Definition 6, showing that the proposed method provides more reliable class-specific predictions and outperforms other methods in minimizing false positives.

Fig. 6 shows the performance of the proposed method based on recall, as defined in Definition 7, demonstrating proposed method superior ability to correctly identify relevant instances compared to competing methods.

5.4. Parameter selection

In the proposed method, a hyperparameter was introduced to determine the number of criteria (i.e., reconstructed representations) to be included in the decision matrix for each dataset. This parameter was empirically set to 0.2 based on preliminary experiments. Specifically, the reconstruction process was terminated when the difference between two successive reconstructed matrices fell below 20% of the difference observed between the first two reconstructed matrices.

The selection of 0.2 as the threshold was justified by the observation that lower values had a negligible impact on the final outcome. This is primarily because the weighting coefficients in the decision matrix

Table 3
The Wilcoxon test results based on classification accuracy.

Accuracy PCA-MCDM against							
Dataset	AEFS [68]	DGUFS [69]	EFS-FMCDM [24]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]
Colon	0.0002(+)	0.0003(+)	0.0002(+)	0.0002(+)	0.0003(+)	0.0450(+)	0.0002(+)
Gulub	0.0036(+)	0.0028(+)	0.0090(+)	0.0113(+)	0.0113(+)	0.0046(+)	0.0028(+)
Khan	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.0028(+)
Nci60	0.0887(=)	0.0002(+)	0.0010(+)	0.0003(+)	0.0002(+)	0.1613(=)	0.0140(+)
Shipp	0.0006(+)	0.0002(+)	0.0013(+)	0.0013(+)	0.0002(+)	0.0002(+)	0.0002(+)
Subramanian	0.0151(+)	0.0125(+)	0.0448(+)	0.0025(+)	0.0004(+)	0.2120(=)	0.1978(=)
Sorlie	0.7335(=)	0.0008(+)	0.2897(=)	0.0006(+)	0.0006(+)	0.0036(+)	0.0017(+)
Su	0.7042(=)	0.0002(+)	0.0013(+)	0.0008(+)	0.0002(+)	0.0058(+)	0.0172(+)
Lung	0.0017(+)	0.0073(+)	0.0046(+)	0.0812(=)	0.0013(+)	0.0013(+)	0.0140(+)
Corel	1.0000(=)	0.0002(+)	0.0211(+)	0.0017(+)	0.1041(=)	0.0140(+)	0.0058(+)
Madelon	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.3447(=)	0.0002(+)	0.0002(+)
Usps	0.0022(+)	0.2730(=)	0.3847(=)	0.0091(+)	0.7337(=)	0.0003(+)	0.0539(=)

Table 4
The Wilcoxon test results based on classification f-measure.

F-measure PCA-MCDM against							
Dataset	AEFS [68]	DGUFS [69]	EFS-FMCDM [24]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]
Colon	0.0002(+)	0.0004(+)	0.0003 (+)	0.0002(+)	0.0002(+)	0.0376(+)	0.0002(+)
Gulub	0.0036(+)	0.0005(+)	0.0091 (+)	0.0113(+)	0.0072(+)	0.0036(+)	0.0029(+)
Khan	0.0002(+)	0.0002(+)	0.00018 (+)	0.0003(+)	0.0002(+)	0.0002(+)	0.0090(+)
Nci60	0.2122(=)	0.0002(+)	0.0376 (+)	0.0036(+)	0.0002(+)	1.0000(=)	0.6775(=)
Shipp	0.0768(+)	0.0002(+)	0.0010 (+)	0.0036(+)	0.0002(+)	0.0002(+)	0.0002(+)
Subramanian	0.0005(+)	0.0889(=)	0.0028 (+)	0.0028(+)	0.0376(+)	0.4726(=)	0.9698(=)
Sorlie	0.3075(=)	0.0007(+)	0.7913 (=)	0.0017(+)	0.0013(+)	0.0057(+)	0.0017(+)
Su	0.5693(=)	0.0002(+)	0.0010 (+)	0.0005(+)	0.0002(+)	0.0036(+)	0.0113(+)
Lung	0.0017(+)	0.0022(+)	0.0091 (+)	0.0211(+)	0.0010(+)	0.0001(+)	0.0140(+)
Corel	1.0000(=)	0.0002(+)	0.01133(+)	0.0017(+)	0.0889(=)	0.0113(+)	0.0045(+)
Madelon	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.3447(=)	0.0002(+)	0.0002(+)
Usps	0.0022(+)	0.2730(=)	0.3846(=)	0.0091(+)	0.7337(=)	0.0003(+)	0.0756(+)

Table 5
The Wilcoxon test results based on classification precision.

Precision PCA-MCDM against							
Dataset	AEFS [68]	DGUFS [69]	EFS-FMCDM [24]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]
Colon	0.0028(+)	0.2730(=)	0.4274(=)	0.0113(+)	0.7913(=)	0.0003(+)	0.0757(+)
Gulub	1.0000(=)	0.0002(+)	0.0113(+)	0.0017(+)	0.0890(=)	0.0173(+)	0.0058(+)
Khan	0.0022(+)	0.0005(+)	0.0046(+)	0.0058(+)	0.0073(+)	0.0036(+)	0.0006(+)
Nci60	0.0006(+)	0.0890(=)	0.0028(+)	0.0028(+)	0.0376(+)	0.4727(=)	0.9698(=)
Shipp	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.3447(=)	0.0002(+)	0.0002(+)
Subramanian	0.5702(=)	0.0002(+)	0.0010(+)	0.0006(+)	0.0002(+)	0.0036(+)	0.0091(+)
Sorlie	0.0003(+)	0.0002(+)	0.0017(+)	0.0008(+)	0.0002(+)	0.0002(+)	0.0002(+)
Su	0.3075(=)	0.0013(+)	0.5205(=)	0.0010(+)	0.0013(+)	0.0028(+)	0.0017(+)
Lung	0.0002(+)	0.0002(+)	0.0002(+)	0.0003(+)	0.0002(+)	0.0002(+)	0.0006(+)
Corel	0.0002(+)	0.0006(+)	0.0006(+)	0.0002(+)	0.0002(+)	0.0017(+)	0.0002(+)
Madelon	0.0017(+)	0.0022(+)	0.0257(+)	0.2123(=)	0.0010(+)	0.0010(+)	0.0173(+)
Usps	0.4726(=)	0.0002(+)	0.1041(=)	0.0539(=)	0.0004(+)	0.5708(=)	0.6776(=)

Table 6
The Wilcoxon test results based on classification Recall.

Recall PCA-MCDM against							
Dataset	AEFS [68]	DGUFS [69]	EFS-FMCDM [24]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]
Colon	0.0002(+)	0.0004(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.5205(=)	0.0006(+)
Gulub	0.0036(+)	0.0028(+)	0.0173(+)	0.0173(+)	0.0091(+)	0.0058(+)	0.0028(+)
Khan	0.0002(+)	0.0002(+)	0.0002(+)	0.0003(+)	0.0002(+)	0.0002(+)	0.0209(+)
Nci60	0.0757(=)	0.0002(+)	0.0013(+)	0.0017(+)	0.0002(+)	0.3447(=)	0.6231(=)
Shipp	0.0028(+)	0.0002(+)	0.0004(+)	0.0211(+)	0.0002(+)	0.0002(+)	0.0002(+)
Subramanian	0.0006(+)	0.0211(+)	0.0006(+)	0.0036(+)	0.0073(+)	0.6232(=)	1.0000(=)
Sorlie	0.4277(=)	0.0008(+)	0.9097(=)	0.0028(+)	0.0013(+)	0.0113(+)	0.0073(+)
Su	0.6227(=)	0.0002(+)	0.0010(+)	0.0013(+)	0.0002(+)	0.0036(+)	0.0211(+)
Lung	0.0010(+)	0.0017(+)	0.0046(+)	0.0046(+)	0.0010(+)	0.0006(+)	0.0046(+)
Corel	0.9696(=)	0.0002(+)	0.0140(+)	0.0017(+)	0.1212(=)	0.0113(+)	0.0046(+)
Madelon	0.0002(+)	0.0002(+)	0.0002(+)	0.0002(+)	0.3447(=)	0.0002(+)	0.0002(+)
Usps	0.0022(+)	0.2730(=)	0.3447(=)	0.0073(+)	0.7337(=)	0.0002(+)	0.0456(+)

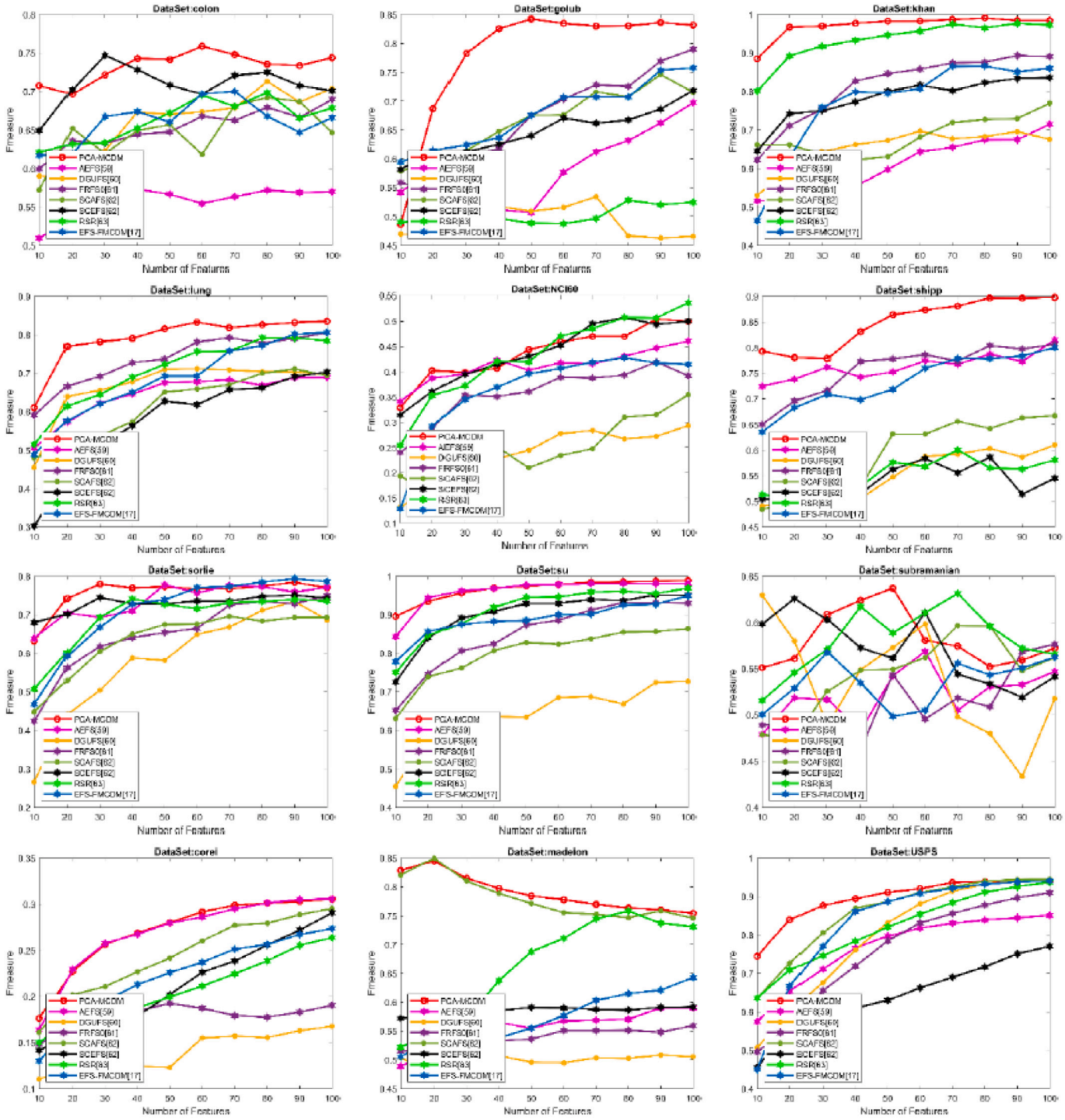


Fig. 4. The results based on classification F-measure.

are derived from the eigenvalues of the covariance matrix. As the magnitudes of the eigenvalues decrease rapidly and often approach zero, incorporating additional criteria beyond a certain point contributes minimally to the feature selection process. Consequently, increasing the number of criteria—for instance, from 20 to 30—did not yield meaningful improvements, as the influence of the additional components was marginal.

Conversely, increasing the threshold value—thereby allowing fewer criteria—was found to degrade the algorithm's performance, as it limited the representational capacity of the decision matrix. This decline in performance associated with higher threshold values is quantitatively illustrated in Fig. 7.

5.5. Discussion

The results demonstrate that the proposed PCA-MCDM-based FS method outperforms traditional methods in terms of accuracy, F-measure, recall, and precision across various datasets. This superior performance highlights the ability of the proposed method to effectively identify diverse and informative features that contribute to downstream

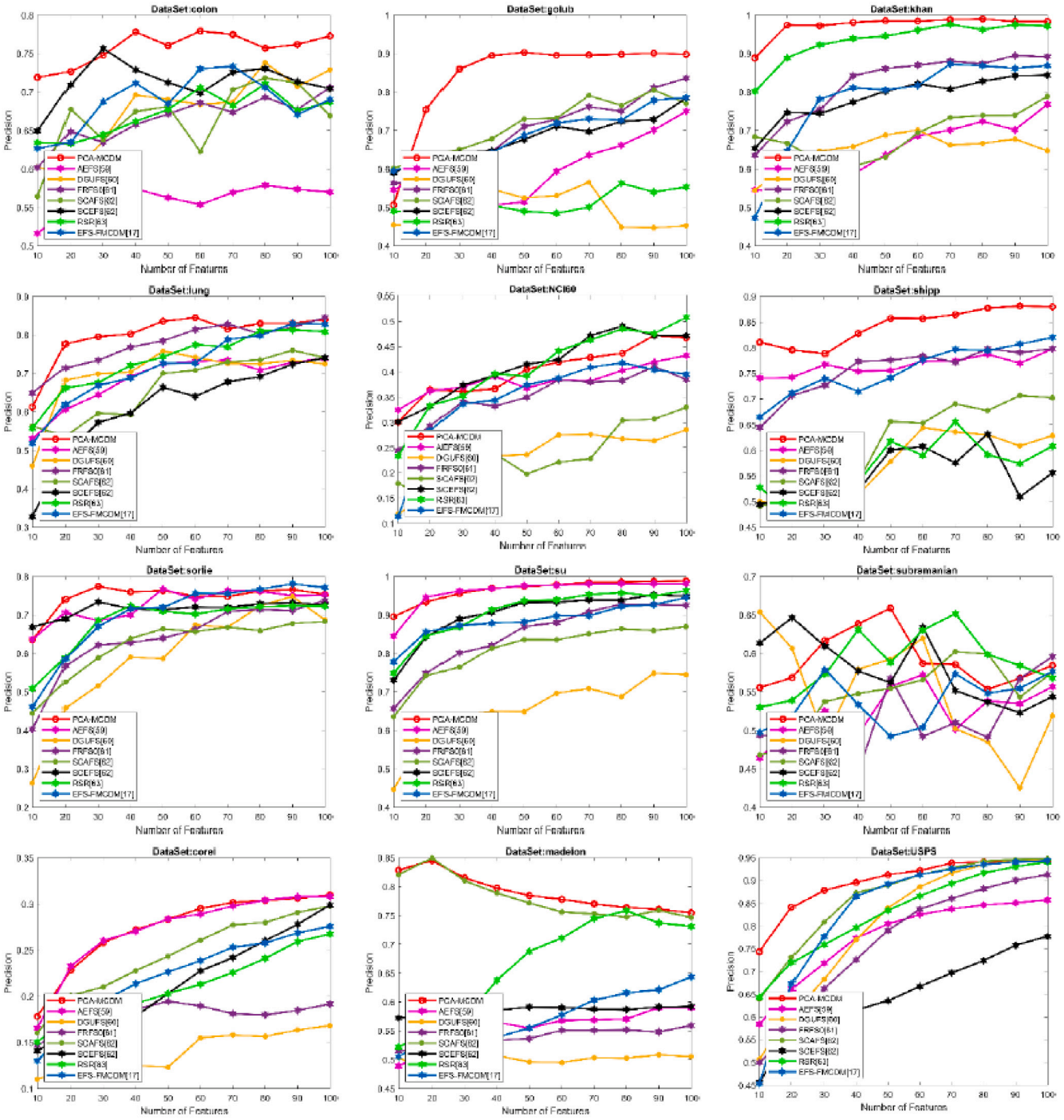


Fig. 5. The results based on classification Precision.

tasks such as classification and clustering. By combining the variance-preserving properties of PCA with the decision-making capability of MOORA, the proposed approach balances the trade-off between capturing significant data structure and maintaining feature diversity, resulting in a feature set that is both representative and non-redundant.

The primary objective of this study was to develop an interpretable, unsupervised FS method that maintains the strengths of PCA while preserving original feature identity. The results presented in Section 5.4 support the effectiveness of our PCA-MCDM fusion method across a variety of datasets, including gene expression, text, and image data. Compared to traditional PCA-based techniques, our method uniquely avoids transforming original features into non-interpretable components. Instead, it uses reconstruction error to assess feature relevance and ranks features via a robust multi-criteria framework. This design aligns well with the demands of real-world applications, particularly in domains like bioinformatics, where domain experts need to trace model decisions back to specific genes or biomarkers. When compared to deep learning-based feature selection methods such as AEFS, our method shows competitive or superior performance in many cases, especially when fewer features are selected. While AEFS benefits from representation learning, it also requires more complex model tuning and infrastructure. Our approach, by contrast, is lightweight and easier

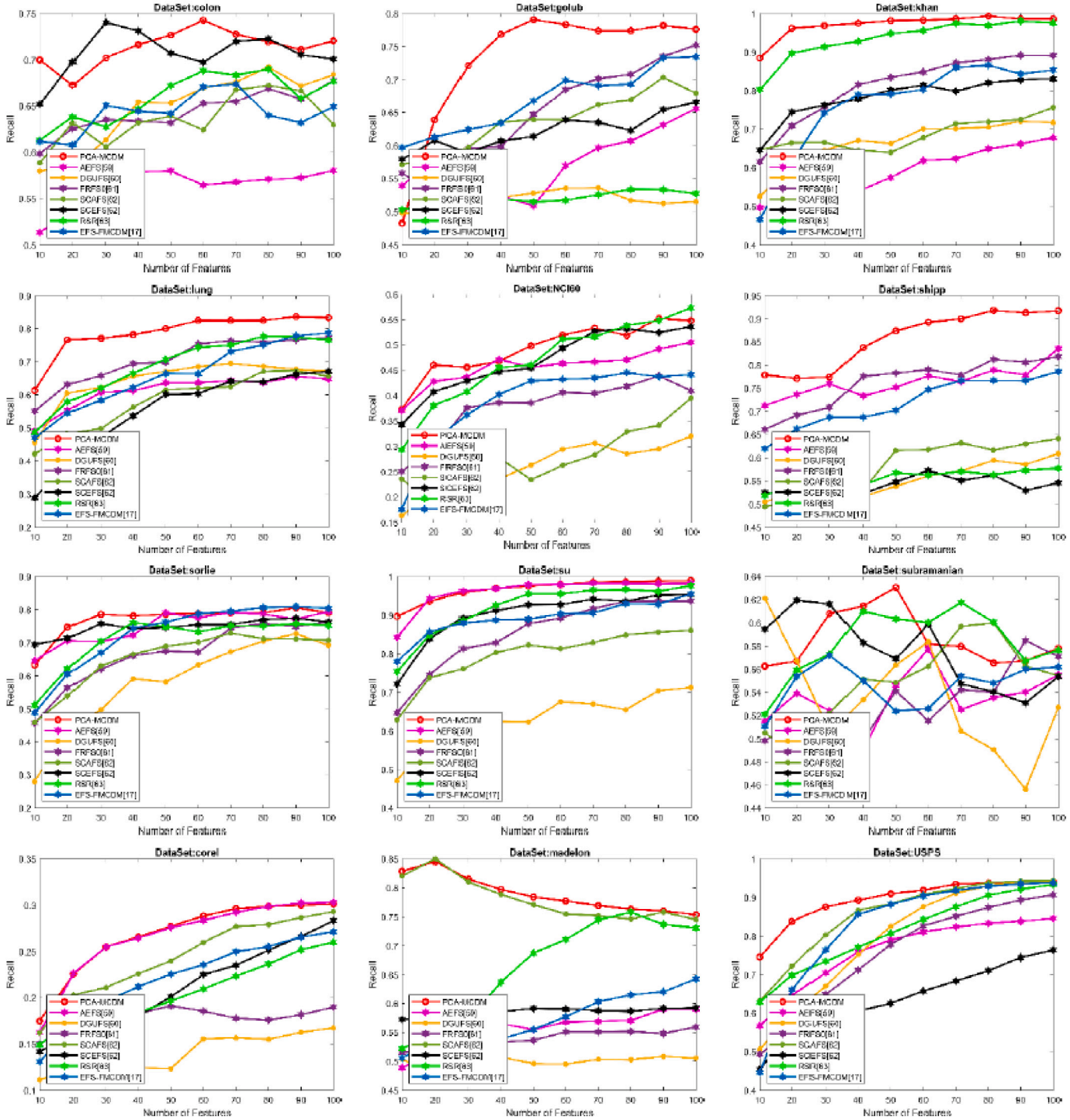


Fig. 6. The results based on classification Recall.

to deploy across small to medium-scale problems. Additionally, compared to filter methods such as ReliefF or statistical tests (e.g., t-test, chi-square), our fusion framework accounts for both global variance structure (via PCA) and multi-criteria evaluation (via MOORA), providing a more balanced and robust feature selection strategy. This positions our work as a bridge between variance-based methods and decision-theoretic approaches in unsupervised learning. From a theoretical perspective, this study contributes to the growing literature on interpretable machine learning by demonstrating how reconstruction error—traditionally used in compression or anomaly detection—can guide feature relevance in a non-supervised context. Furthermore, the integration of MCDM methods into feature selection offers a flexible pathway for extending this framework with other decision strategies (e.g., VIKOR, TOPSIS) in future work. While the method is based on deterministic matrix operations (e.g., PCA decomposition, MOORA ranking), its runtime can be affected by high-dimensional feature spaces and the lack of low-level optimization in current implementations. Deep learning methods like AEFS, although seemingly more complex, benefit from optimized libraries and GPU execution, leading to faster runtime in certain cases. This does not undermine the practicality of

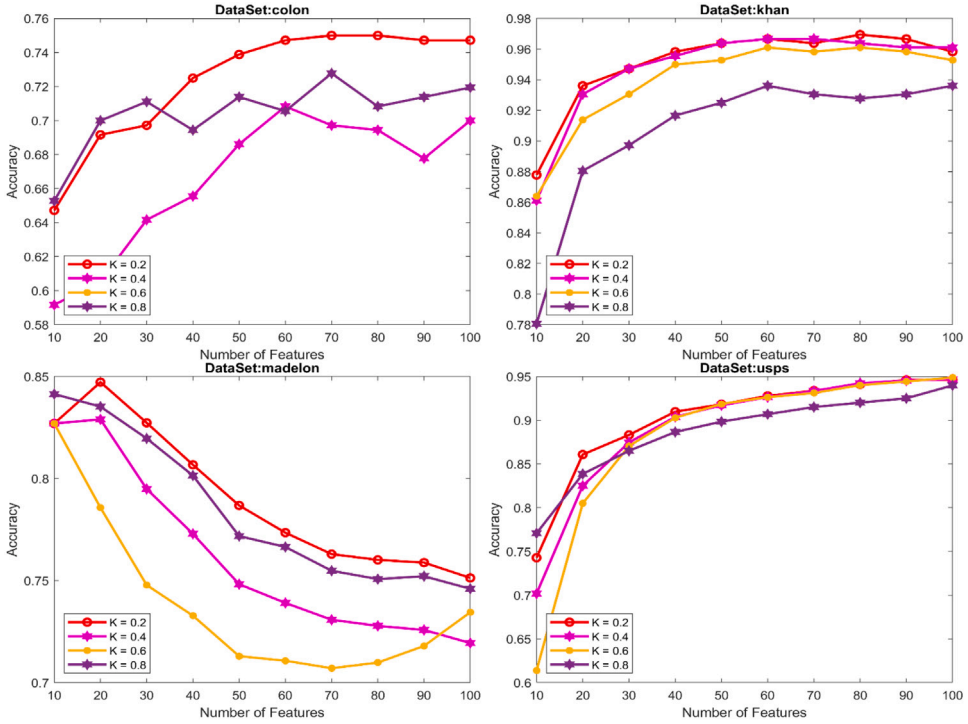


Fig. 7. Parameter selection result.

Table 7
The overall win/tie/loss based on the Wilcoxon test.

	Overall						
Metric	AEFS [68]	DGUFS [69]	EFS-FMCDM [24]	FRFSO [70]	SCAFS [71]	SCEFS [71]	RSR [72]
Accuracy	8/4/0	11/1/0	10/2/0	11/1/0	9/3/0	10/2/0	10/2/0
F-measure	8/4/0	10/2/0	10/2/0	12/0/0	9/3/0	10/2/0	10/2/0
Precision	8/4/0	10/2/0	9/3/0	10/2/0	9/3/0	10/2/0	10/2/0
Recall	8/4/0	11/1/0	10/2/0	12/0/0	9/3/0	9/3/0	10/2/0

our approach, as it remains simpler to configure, fully interpretable, and highly transferable to various data types. Future work will explore optimized implementations and dimensionality-aware variants to further enhance runtime efficiency. The proposed FS framework is primarily designed for high-dimensional structured datasets, such as gene expression data or text-term matrices. However, the modular design of the method allows for potential extension to semi-structured (e.g., XML-based sensor logs) and unstructured datasets (e.g., raw text or image data), assuming that suitable numeric feature representations (such as TF-IDF vectors, word embeddings, or image descriptors) are available. PCA can operate on these representations to extract principal components, while the MOORA-based ranking can be applied to reconstruction errors or other feature relevance scores.

Regarding scalability, both PCA and MOORA are computationally efficient for moderate-scale datasets. For larger datasets, techniques such as randomized or incremental PCA, along with distributed implementations of MCDM frameworks, could be integrated. These adaptations would allow the method to scale effectively in big data environments. Future work will explore such optimizations and investigate performance trade-offs across data types and volumes.

5.6. Limitations

While the proposed PCA-MCDM fusion method has shown strong performance across various high-dimensional datasets, several important limitations must be acknowledged to provide a balanced view of its applicability and future potential:

· **Scalability to Ultra-High-Dimensional Data:** Although the method is computationally efficient compared to deep learning models, it may face runtime challenges when applied to ultra-high-dimensional datasets (e.g., those with more than 20,000 features). This is primarily due to the computational cost of full PCA decomposition. Moreover, as feature dimensionality increases, memory constraints and matrix operations may become bottlenecks. Future work could explore more scalable variants such as randomized or incremental PCA to address this limitation.

- **Dependency on PCA Orthogonality:** The method assumes that principal components are linearly uncorrelated, which may not fully capture complex nonlinear relationships among features. In datasets where such nonlinear dependencies dominate, the reconstruction-error-based feature relevance measure may be less informative. Extending the framework to incorporate kernel PCA or nonlinear dimensionality reduction techniques could enhance its generalizability.
- **Lack of Confidence Quantification:** The evaluation in this study is based on average classification accuracy across different feature subset sizes. However, no confidence intervals or statistical significance tests were included. Future work could integrate bootstrapping or cross-validation with confidence bounds to strengthen the statistical reliability of the results.
- **Limited Validation on Unstructured Data:** The method was primarily validated on structured, tabular datasets. While conceptually applicable to semi-structured or unstructured data (e.g., raw text, sensor signals, or image features), such applications would

require careful feature engineering or embedding techniques. Future studies should explore the method's performance in those contexts.

- **Absence of Adaptive Feature Subset Selection:** The approach currently evaluates performance across a predefined range of feature subset sizes. It does not automatically determine an optimal number of features to retain. Incorporating adaptive selection mechanisms, such as stability-based criteria or inflection-point detection, could improve usability in practical applications.

Despite these limitations, the proposed method offers a transparent, flexible, and model-agnostic solution for unsupervised feature selection. Its simplicity and interpretability make it particularly valuable in domains where feature traceability is essential. Future work will aim to address the above challenges and explore hybrid designs that integrate more advanced ranking or reduction strategies.

6. Conclusion and future works

This study presents a novel hybrid approach for unsupervised feature selection that integrates PCA with the MOORA method from MCDM. While PCA is a widely used technique for dimensionality reduction, it often fails to preserve the interpretability of original features—a crucial requirement in many real-world applications such as gene expression analysis, text mining, and medical diagnostics. By leveraging reconstruction error from PCA and applying MOORA for feature ranking, our method bridges the gap between statistical variance capture and interpretable feature selection.

From a scientific standpoint, the proposed approach contributes to the ongoing discourse on interpretable machine learning by offering a technique that maintains data fidelity while enhancing decision transparency. Unlike traditional PCA-based methods that transform features into unrecognizable components, our method retains the original feature space—making it suitable for high-stakes domains where domain expertise and traceability are critical.

From a societal perspective, this method supports more accessible and transparent data analysis, particularly in fields like bioinformatics and healthcare, where understanding which features (e.g., genes, symptoms, terms) influence model predictions can guide real-world interventions, policy, or clinical decisions. The key insights and contributions of this research are as follows:

- We propose a novel way to reinterpret PCA—not as a feature extractor, but as a guiding tool for selecting original features based on reconstruction error.
- We integrate this PCA-based relevance assessment into the MOORA framework to allow for robust, criteria-based feature ranking in an unsupervised setting.
- We demonstrate the method's effectiveness across diverse datasets, showing consistent improvements in classification performance with fewer selected features.
- We highlight its generalizability and practical value by applying it to multiple data types beyond bioinformatics, thus extending its relevance.

In summary, this work offers a new lens through which PCA and MCDM techniques can be combined to address critical challenges in unsupervised feature selection. Future work will focus on optimizing runtime, exploring alternative distance metrics, and expanding applications to unstructured and semi-structured data sources. The simplicity, transparency, and adaptability of the proposed method position it as a valuable contribution to interpretable and efficient machine learning.

The results demonstrate that the proposed method is both effective and computationally efficient. Future research will focus on extending its applicability to other feature selection contexts, including multi-label and semi-supervised classification. Moreover, its performance will be further evaluated across diverse machine learning applications and scenarios to assess its generalizability and robustness.

CRediT authorship contribution statement

Mohsen Habibollahi: Writing – original draft, Methodology. **Amin Hashemi:** Investigation, Conceptualization. **Mohammad Bagher Dowlatshahi:** Validation, Resources. **Marjan Kuchaki Rafsanjani:** Supervision, Data curation. **Varsha Arya:** Resources, Formal analysis. **Brij B. Gupta:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data generated during the study are subject to a data sharing mandate and available in a few public repositories. All used data are cited in text.

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